

Locality of $\text{Tr}(\phi^3)$ tree amplitudes from cyclic hidden zeros

Part II: the fat-zero generalisation

Abstract

Part I established locality through $n = 9$ by combining a single asymptotic “kill” on one hidden 1-zero with a per- n Laurent cascade. Here we promote the 1-zero to a k -zero (“fat zero”): the simultaneous imposition of the hidden-zero relations on k consecutive boundaries. A k -zero kills strictly more nonlocal coefficients than the 1-zero. We derive its constraint surface and classify, by a rate-vector analysis, all admissible asymptotic limits in which the special chord diverges—i.e. the criteria A , B , C promised in the title. The geometric shadow of these limits is the construction of a $(k+2)$ -gon on top of the length- k enclosing chord; we show there are three flavours of completion (special, bare, and an internal *column pair*). These criteria are the correct objects; the open problem is to assemble them, across all zeros, into an all- n proof. The Step-2 coefficient equations, now classified for the 1-, 2-, and 3-zeros, are the tool for that assembly and are documented in Part III.

1 From one zero to many

We keep the conventions of Part I. The n -gon has vertices $1, \dots, n$; a chord X_{ij} ($1 \leq i < j \leq n$, $|j - i| \neq 1$, $(i, j) \neq (1, n)$) is a variable; an edge $X_{m, m+1}$ is a leg and equals 0. The ansatz is $B_n = \sum_{|M|=n-3} a_M / \prod_{c \in M} X_c$ over multisets M , and locality is the statement $a_M = 0$ whenever M contains a crossing pair.

A cyclic hidden 1-zero imposes the $n - 3$ relations $c_{1j} = 0$, where

$$c_{ij} = X_{ij} + X_{i+1, j+1} - X_{i, j+1} - X_{i+1, j}. \quad (1)$$

On $\mathcal{Z}_{1,3}$ these read $X_{2k} = X_{1k} - X_{13}$ and $X_{2n} = -X_{13}$; the special chord is X_{13} and the bare chord is X_{2n} .

Definition 1 (k -zero). For $1 \leq k \leq n - 3$, the k -zero based at $\{1, \dots, k+1\}$ is the simultaneous imposition

$$c_{ij} = 0, \quad i = 1, \dots, k, \quad j = k + 2, \dots, n - 1. \quad (2)$$

It collects the hidden-zero relations on the k consecutive boundaries $(1, 2), (2, 3), \dots, (k, k+1)$. For $k = 1$ it is the ordinary 1-zero. There are $k(n - k - 2)$ relations, and they cut out a linear surface $\mathcal{Z}^{(k)}$ that we now describe explicitly.

2 The constraint surface of a k -zero

Lemma 1 (Telescoping). On $\mathcal{Z}^{(k)}$, with the leg convention $X_{m, m+1} = 0$,

$$X_{k+1, j} = X_{1j} - X_{1, k+2}, \quad j = k + 3, \dots, n - 1; \quad X_{k+1, n} = -X_{1, k+2}, \quad (3)$$

$$X_{ij} = X_{k+1, j} + X_{i, k+2}, \quad i = 2, \dots, k, \quad j = k + 3, \dots, n. \quad (4)$$

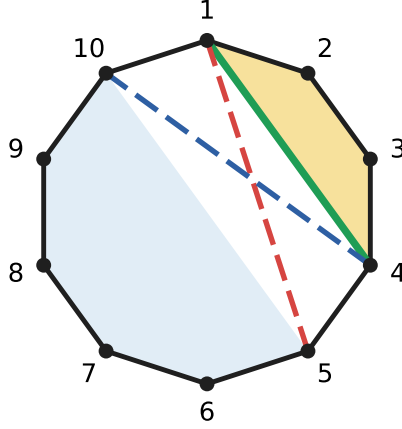


Figure 1: Geometry of a k -zero ($n = 10$, $k = 3$). Near block $N = \{1, 2, 3, 4\}$ (gold), far block $F = \{5, \dots, 10\}$ (blue). Green: the length- k enclosing chord $X_{1,4}$. Red dashed: the special $X_{1,5}$. Blue dashed: the bare $X_{4,10}$. Born-free chords live inside N or inside F .

Proof. Sum (1) over the full block of rows $i = 1, \dots, k$ at fixed j . The two inner sums telescope and leave $\sum_{i=1}^k c_{ij} = (X_{1j} - X_{k+1,j}) - (X_{1,j+1} - X_{k+1,j+1}) = 0$, so $X_{k+1,j} - X_{1j}$ is independent of j . Evaluating at $j = k+2$ (where $X_{k+1,k+2} = 0$) gives the constant $-X_{1,k+2}$, which is (3); at $j = n$ one has $X_{1n} = 0$, giving the bare relation. Summing instead the partial block $i' = i, \dots, k$ gives $X_{ij} - X_{i,j+1} = X_{k+1,j} - X_{k+1,j+1}$; telescoping this in j from $k+2$ (where $X_{k+1,k+2} = 0$) yields (4). $\square$

Three structures organise $\mathcal{Z}^{(k)}$ (Figure 1).

- The **near block** $N = \{1, \dots, k+1\}$ and the **far block** $F = \{k+2, \dots, n\}$. Every chord internal to N or to F is absent from (2): these are the *born-free* chords, untouched by the k -zero.
- The length- k **enclosing chord** $X_{1,k+1}$ closes the k consecutive boundaries into the $(k+1)$ -gon N . It is born-free.
- The **special** $X_{1,k+2}$ and the **bare** $X_{k+1,n}$, the exact analogues of X_{13} and X_{2n} : each bridges N to F across one boundary of N .

All remaining constrained chords are *bridges* from a near vertex to a far vertex, grouped into *columns* indexed by the far endpoint j : column j is $\{X_{1j}, X_{2j}, \dots, X_{k+1,j}\}$. By Lemma 1 a column is fixed by the single free choice X_{1j} together with the shared offsets $X_{1,k+2}$ (special) and $X_{i,k+2}$ ($i = 2, \dots, k$). A convenient independent coordinate system on $\mathcal{Z}^{(k)}$ is therefore

$$\underbrace{\{X_{1,k+2}\}}_{\text{special } s} \cup \underbrace{\{X_{i,k+2} : 2 \leq i \leq k\}}_{\text{column-}(k+2) \text{ offsets}} \cup \underbrace{\{X_{1\ell} : k+3 \leq \ell \leq n-1\}}_{\text{one per far column}} \cup \{\text{born-free}\}, \quad (5)$$

of which the $n-3$ bridge coordinates determine every constrained chord.

3 The kill: a rate-vector classification

The kill mechanism is the Laurent argument of Part I. A one-parameter limit on $\mathcal{Z}^{(k)}$ sends some chords to infinity; the terms whose denominators remain finite form a vanishing Laurent polynomial in the finite coordinates, so $a_M = 0$ for every M supported on the finite set $\mathcal{F}$.

What is special about the k -zero is that the surface $\mathcal{Z}^{(k)}$ admits a rich family of such limits, and the constraints (3)–(4) couple the chords, so we are *not* free to hold an arbitrary subset finite. We parametrise the limits exactly. Fix a divergent parameter $s \rightarrow \infty$ (the special) and assign each remaining bridge coordinate a leading rate:

$$X_{i,k+2} \sim \beta_i s \ (i = 2, \dots, k), \quad X_{1\ell} = u_\ell \sim \alpha_\ell s \ (\ell = k+3, \dots, n-1), \quad (6)$$

with $\alpha_\ell, \beta_i \in \mathbb{R}$ and equality of leading coefficients understood. Each chord on $\mathcal{Z}^{(k)}$ then has a definite rate, read off from (3)–(4):

chord	rate	finite iff
$X_{1\ell}$	$\alpha_\ell s$	$\alpha_\ell = 0$
$X_{k+1,\ell} = u_\ell - s$	$(\alpha_\ell - 1) s$	$\alpha_\ell = 1$
$X_{i\ell} = u_\ell - s + X_{i,k+2}$	$(\alpha_\ell + \beta_i - 1) s$	$\alpha_\ell + \beta_i = 1$
$X_{i,k+2}$	$\beta_i s$	$\beta_i = 0$
$X_{in} = X_{i,k+2} - s$	$(\beta_i - 1) s$	$\beta_i = 1$
$X_{1,k+2} = s$	s	<i>never</i>
$X_{k+1,n} = -s$	$-s$	<i>never</i>

(7)

Each rate vector $(\alpha_\ell; \beta_i)$ thus determines an admissible finite-set

$$\mathcal{F}(\alpha, \beta) = \{\text{born-free}\} \cup \{\text{the chords whose “finite iff” clause in (7) holds}\}. \quad (8)$$

Theorem 1 (Admissible finite-sets). *Every admissible finite-set on $\mathcal{Z}^{(k)}$ based at $\{1, \dots, k+1\}$ has the form $\mathcal{F}(\alpha, \beta)$ for some $(\alpha_\ell; \beta_i) \in \mathbb{R}^{(n-k-3)+(k-1)}$. The special $X_{1,k+2}$ and the bare $X_{k+1,n}$ are absent from every $\mathcal{F}(\alpha, \beta)$. Consequently*

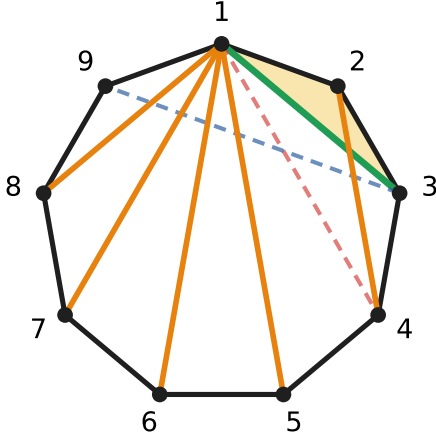
$$a_M = 0 \quad \text{whenever } M \subseteq \mathcal{F}(\alpha, \beta) \text{ for some } (\alpha, \beta).$$

Proof. Every limit on $\mathcal{Z}^{(k)}$ is determined by the leading scaling of the independent coordinates (5); (6) captures this for the bridges, the born-free coordinates are held finite. The rate of each remaining chord is given by (3)–(4), and is finite precisely under the listed condition. The Laurent-polynomial vanishing on the finite coordinates then forces $a_M = 0$ for $M \subseteq \mathcal{F}$. The special and bare carry an uncancellable bare power of s . $\square$

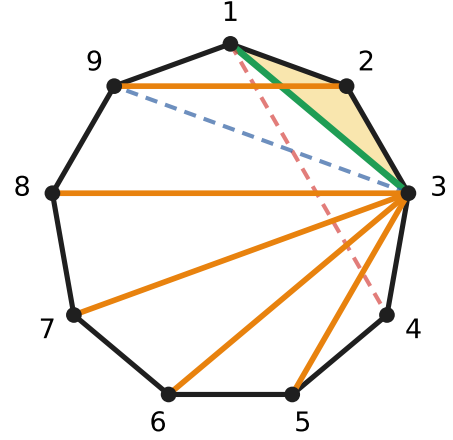
The named regimes are the three structurally distinct ways of fixing the rate vector:

(A) 1-side	all $\beta_i = 0$ and all $\alpha_\ell = 0$.	$\mathcal{F}_A = \{X_{i,k+2} : 2 \leq i \leq k\} \cup \{X_{1\ell} : k+3 \leq \ell \leq n-1\}$.
(B) $(k+1)$-side	all $\beta_i = 1$ and all $\alpha_\ell = 1$.	$\mathcal{F}_B = \{X_{in} : 2 \leq i \leq k\} \cup \{X_{k+1,\ell} : k+3 \leq \ell \leq n-1\}$.
(C) mixed	any other consistent rate vector.	per-column slot picks plus offset pairings, as dictated by (7).

A and B are homogeneous: setting all the α 's and β 's to the same value finitises exactly one slot per column. C is everything else—e.g. taking $\beta_i = 0$ but $\alpha_\ell = 1$ for some ℓ makes *both* $X_{k+1,\ell}$ and $X_{i\ell}$ finite, a paired configuration not seen in either homogeneous regime. Figure 3 shows the canonical per-column three slots that organise the C family (one finite chord per column, before pairings); the full C family is the family of all rate vectors compatible with the chosen $\mathcal{F}$ via (7).



Regime A



Regime B

Figure 2: The held-finite bridges (orange) for the two homogeneous regimes, $n = 9$, $k = 2$ (special/bare faint dashed, enclosing chord green). *A* keeps the 1-side $\{X_{24}; X_{15}, X_{16}, X_{17}, X_{18}\}$; *B* keeps the $(k+1)$ -side $\{X_{29}; X_{35}, X_{36}, X_{37}, X_{38}\}$. Both are special cases of Theorem 1.

4 The geometry: $(k+2)$ -gons over the enclosing chord

Theorem 1 has a clean geometric shadow when restricted to multisets that do not cross the enclosing chord.

Proposition 1 ($(k+2)$ -gon survival). *Let M be a multiset using no chord that crosses $X_{1,k+1}$. Then M survives the k -zero based at $\{1, \dots, k+1\}$ if and only if M contains a $(k+2)$ -gon completion of the near block N : exactly one of*

- (i) *the special $X_{1,k+2}$ (closing N into $\{1, \dots, k+2\}$);*
- (ii) *the bare $X_{k+1,n}$ (closing N into $\{n, 1, \dots, k+1\}$);*
- (iii) *a column pair $\{X_{1m}, X_{k+1,m}\}$ for some $m \in \{k+3, \dots, n-1\}$ (closing N into $\{1, \dots, k+1, m\}$).*

See Figure 4. Each completion is the unique way to extend N to a $(k+2)$ -gon across exactly one further polygon boundary.

Proof. Under the no-crossing hypothesis every bridge in M has near endpoint 1 or $k+1$, i.e. is of the form X_{1j} , $X_{k+1,j}$ (including the special $X_{1,k+2}$ and the bare $X_{k+1,n}$). M survives iff no rate vector (α, β) finitises every chord of M . By (7) the conditions imposed by M 's bridges are: $\alpha_\ell = 0$ for each $X_{1\ell} \in M$ with $\ell \in \{k+3, \dots, n-1\}$, $\alpha_\ell = 1$ for each $X_{k+1,\ell} \in M$, and (impossible) clauses for special and bare. The constraint system is inconsistent precisely when one of (i)–(iii) holds: the special or the bare forces an impossible $1=0$ clause; a column pair forces $\alpha_m=0$ and $\alpha_m=1$. Conversely if none of (i)–(iii) holds, the system is consistent (set $\alpha_\ell = 0$ where $X_{1\ell} \in M$, $\alpha_\ell = 1$ where $X_{k+1,\ell} \in M$, any value elsewhere; β_i free), so $M \subseteq \mathcal{F}(\alpha, \beta)$ and is killed. The $(k+2)$ -gons containing N are exactly $\{1, \dots, k+1, m\}$ for $m \in F$, and each is closed by the chord(s) in (i)–(iii). $\square$

The complete (crossing-allowed) survival criterion is the linear-algebra contrapositive of (7): M survives iff the system of rate equations imposed by the chords of M is inconsistent in (α_ℓ, β_i) . Geometrically the simplest non- $(k+2)$ -gon survival modes are pairings of bridges that share a

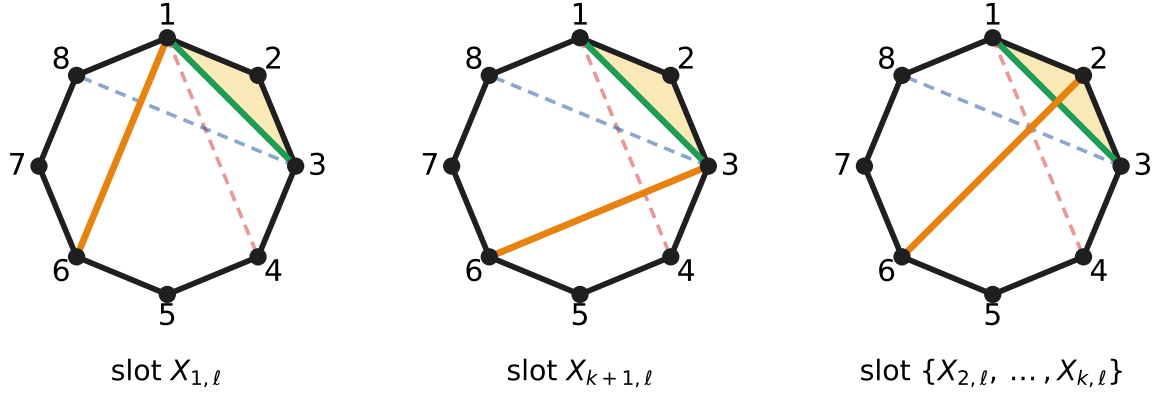


Figure 3: Regime C : the three exclusive *single-chord* slots per far column, here $\ell = 6$, $n = 8$, $k = 2$. Each rate vector (α, β) picks one per column, and—via the offset rates β_i —may simultaneously pair additional interior bridges $X_{i\ell}$ into $\mathcal{F}$.

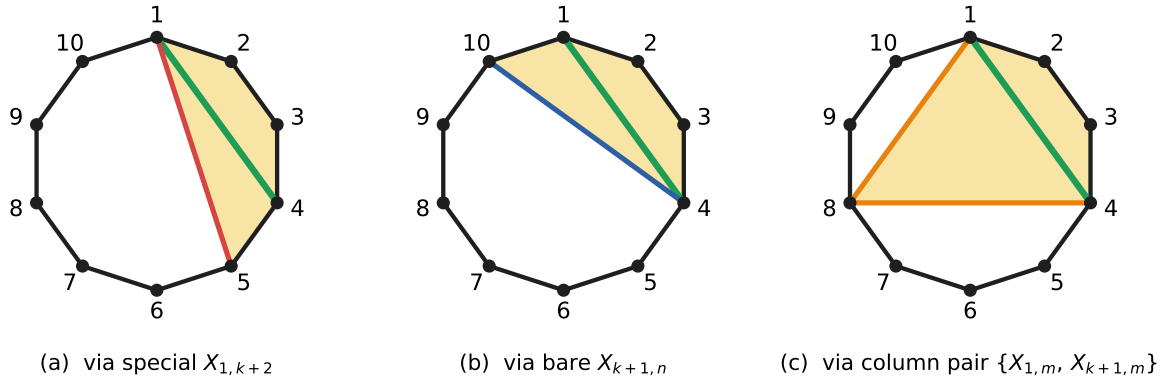


Figure 4: The three $(k+2)$ -gon completions of N ($n = 10$, $k = 3$, $m = 8$ for the column pair). In every non-crossing survivor, exactly one of these occurs.

far vertex from the offsets (e.g. $\{X_{i,k+2}, X_{in}\}$ forcing $\beta_i = 0$ and $\beta_i = 1$), or longer cycles of incompatible rate constraints.

5 Survivors

A triangulation always escapes the k -zero: if the $(k+1)$ -gon over k consecutive edges is self-contained (no chord of the triangulation crosses the enclosing chord $X_{1,k+1}$), then adjoining $X_{1,k+1}$ crosses nothing, so it is already a chord of the triangulation, and—being a chord of a triangulation—bounds a triangle on its far side, supplying a $(k+2)$ -gon completion. By Proposition 1 the triangulation survives.

The conjecture of the program is that, ranging over all k and all cyclic rotations, the union of the k -zero kills removes every non-triangulation, so that exactly the triangulations survive.

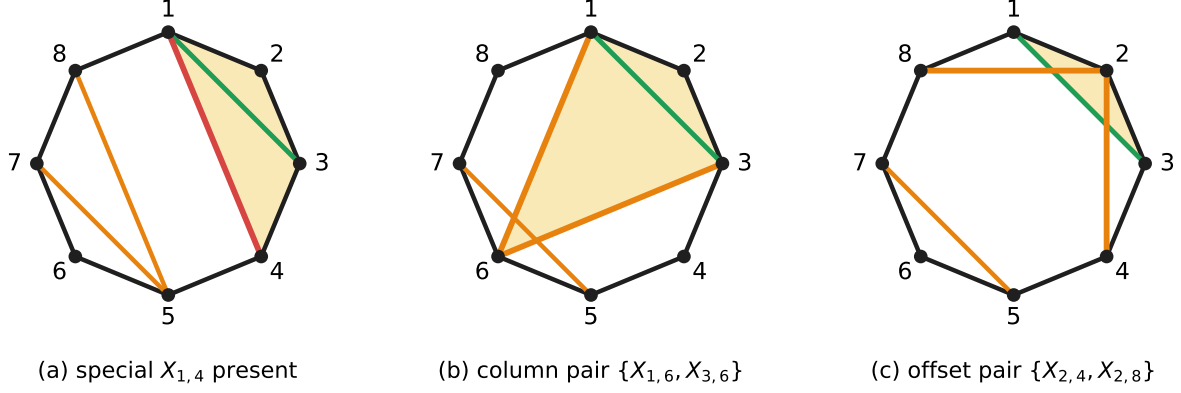


Figure 5: Three families of survivors of the $k = 2$ zero based at $\{1, 2, 3\}$, $n = 8$. (a) The special $X_{1,4}$ is present (the simplest $(k+2)$ -gon completion). (b) A column pair $\{X_{1,6}, X_{3,6}\}$ forces $\alpha_6=0$ and $\alpha_6=1$ simultaneously. (c) An offset pair $\{X_{2,4}, X_{2,8}\}$ forces $\beta_2=0$ and $\beta_2=1$ simultaneously. In every case the rate-equation system is inconsistent, so no admissible $\mathcal{F}$ contains M .

6 Status and the road ahead

Theorem 1 is the corrected statement of the kill: the survival criteria A , B , C are the three structural cases of a single rate-vector classification, derived rigorously from the coupling (3)–(4). The remaining obstacle is structural: a single zero, even read across its full C family, is a single linear projection of the ansatz, and establishing that the *intersection* of all k -zero kills (over all k and all rotations) is precisely the set of triangulations requires reading projections together.

The tool for that assembly is the family of *Step-2 equations*: grouping the vanishing condition on $\mathcal{Z}^{(k)}$ by the number of boundary-propagator factors (equivalently, by the z -scaling weight of Part I) and by the choice of born-free denominator, then clearing denominators and taking the $X \rightarrow 0$ cuts, yields a closed linear system among the surviving coefficients. For the 1-zero these reproduce $a_{13} = a_{2n}$ and the unitarity relations; the 2-zero already contributes new quadratic relations such as $a_{14,14} + a_{3n,3n} - a_{14,3n} = 0$ and the cubic $a_{14,14,14} - a_{14,14,3n} + a_{14,3n,3n} - a_{3n,3n,3n} = 0$ at three boundary propagators; the 3-zero produces the next family. Part III documents the full k -zero Step-2 classification and the linear system it generates.